

◆ What is a File System?

A **file system** is like the **method of organizing and storing files** on a disk (HDD, SSD, USB).

👉 Think of it as the **rules of arrangement** — like how you arrange books on shelves (by subject, size, or author).

◆ FAT32 (File Allocation Table 32)

- **Older file system** (introduced in 1996).
- Widely supported (works with Windows, Linux, macOS, cameras, TVs, game consoles, USB drives).
- **File size limit:** Maximum file size = **4 GB**.
- **Partition size limit:** Maximum = **2 TB**.
- **Security:** No file-level permissions or encryption.

👉 Good for: **USB drives, memory cards, small disks, compatibility with multiple devices.**

◆ NTFS (New Technology File System)

- **Modern file system** used by Windows (since Windows XP).
- **File size limit:** Supports very large files (up to 16 TB and more).
- **Partition size limit:** Supports up to **256 TB**.
- **Security:** Supports permissions (who can read/write a file), encryption, compression.
- **Reliability:** Keeps logs of changes (helps recover after crashes).
- Not as compatible with non-Windows devices (macOS can read NTFS but not write without extra tools).

👉 Good for: **Windows system drives (C:), large hard drives, files bigger than 4 GB.**

◆ Key Differences Table

Feature	FAT32 📁	NTFS 📁
Age	Old (1996)	Newer (1993, improved later)
Max file size	4 GB	Very large (16 TB+)
Max partition size	2 TB	256 TB
Security	No permissions, no encryption	Permissions, encryption, compression
Compatibility	Works everywhere (Windows, Linux, macOS, consoles, cameras)	Mostly Windows (macOS = read-only)
Usage	USB drives, memory cards	Windows system drives, large

✓ In short:

- **FAT32** → Best for **USBs and external drives** if you want them to work everywhere.
- **NTFS** → Best for **Windows system and large storage** (no 4 GB limit).